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Utilizing Smartphone-derived Photogrammetry 3D Model for AI-based Riparian Crack Segmentation and Measurement

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Abstract. In this paper, the authors present a simple free methodology that harnesses the potential of smartphone-derived photogrammetry 3D models for the AI-driven segmentation and quantification of riparian cracks. The approach the authors propose leverages the YOLOv7-seg model to precisely delineate cracks, employing photogrammetry techniques to construct an orthophoto, subsequently enabling the extraction and accurate measurement of crack dimensions. This research underscores the fact that the proposed methodology yields outcomes in both the detection and measurement aspects of riparian cracks. This investigation carries profound implications for the realm of environmental oversight and administration, as it heralds an innovative approach toward the identification and amelioration of issues pertaining to cracks in riparian asphalt surfaces. Through the rigorous analysis of this approach, not only crack detection and measurement but also holds promise for wider applications in the field of environmental assessment. It extends the boundaries of AI-driven methodologies in the context of environmental preservation.

Keywords: Instance Segmentation, Riparian Crack Detection, River Monitoring, Smartphone

1 Introduction

The monitoring of cracks along river banks is a vital aspect of the evaluation of the environmental stabilization of riparian areas. Recently, promising progress has been made in riverbank crack monitoring by using Unmanned Aerial Vehicle (UAV) platforms for artificial intelligence tasks such as instance segmentation [1]. However, high accurate instance segmentation using closer distance has not been thoroughly verified and evaluated. To address this gap, the authors collected several angle-fixed videos along the targeted river bank for selecting the suitable one, and this video was used to be inferred by the You Only Look Once version 7 (YOLOv7) instance segmentation model specifically tailored for crack segmentation, referred to as YOLOv7-seg. In this research, the authors generated a work flow that the asphalt-paved cracks can be extracted and size-measured automatically, and by a free version 3D model processing software, the video-derived frames can be transformed to an orthophoto, which these results can be important data accumulation for the construction ecosystem digital-twin in the future.

2 Study site

Figure 1 illustrates the study site, one top-faced embankment in the Asahi River, a first-class river in the Asahi River system in the Okayama City, Okayama Prefecture. In this area, the riparian asphalt pavements are suffering from the cracks in the road, i.e., the one in the yellow square. One of the reason choosing this area is derived from the amount and multiple species of the cracks (i.e., longitudinal, lateral). Another reason is that, there are some buildings near this area with some residence living there, so the safety of top-surface embankment in this area is necessary to be monitored frequently before the flood period. And in the case of that the cars are allowed to move in this area, the data collection was operated by the researchers during the safety situation (i.e., car-free).

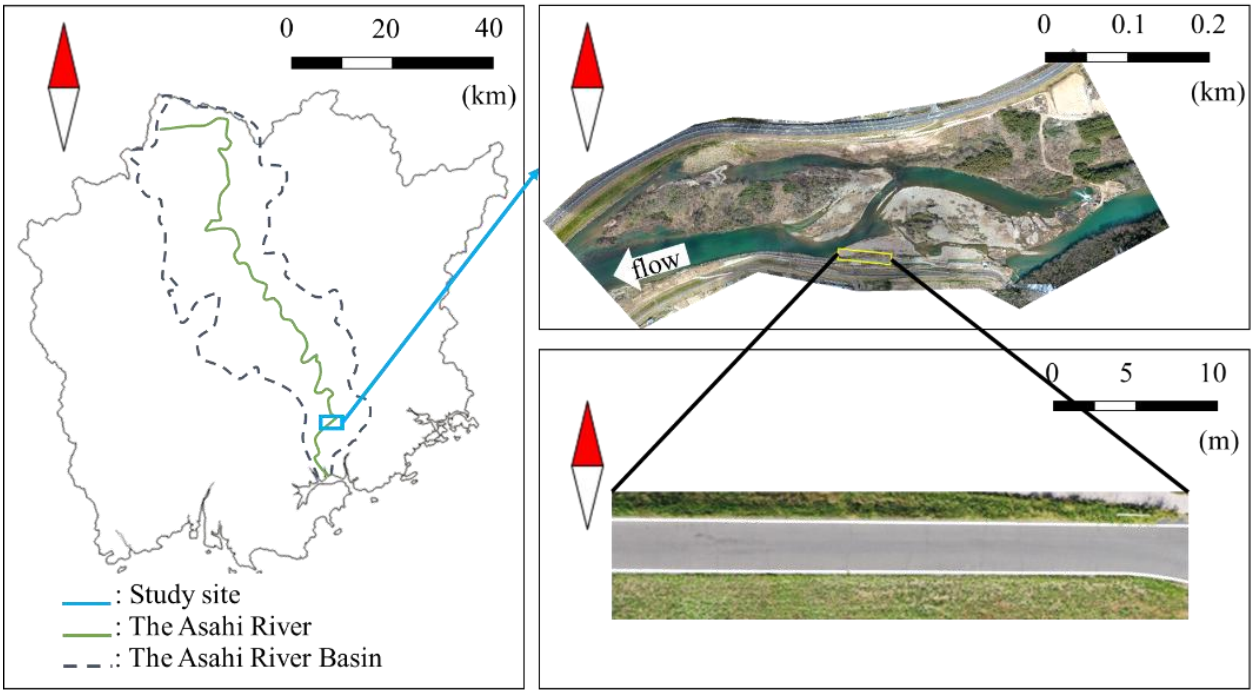

Figure 1 Process of the data collection and the workflow .

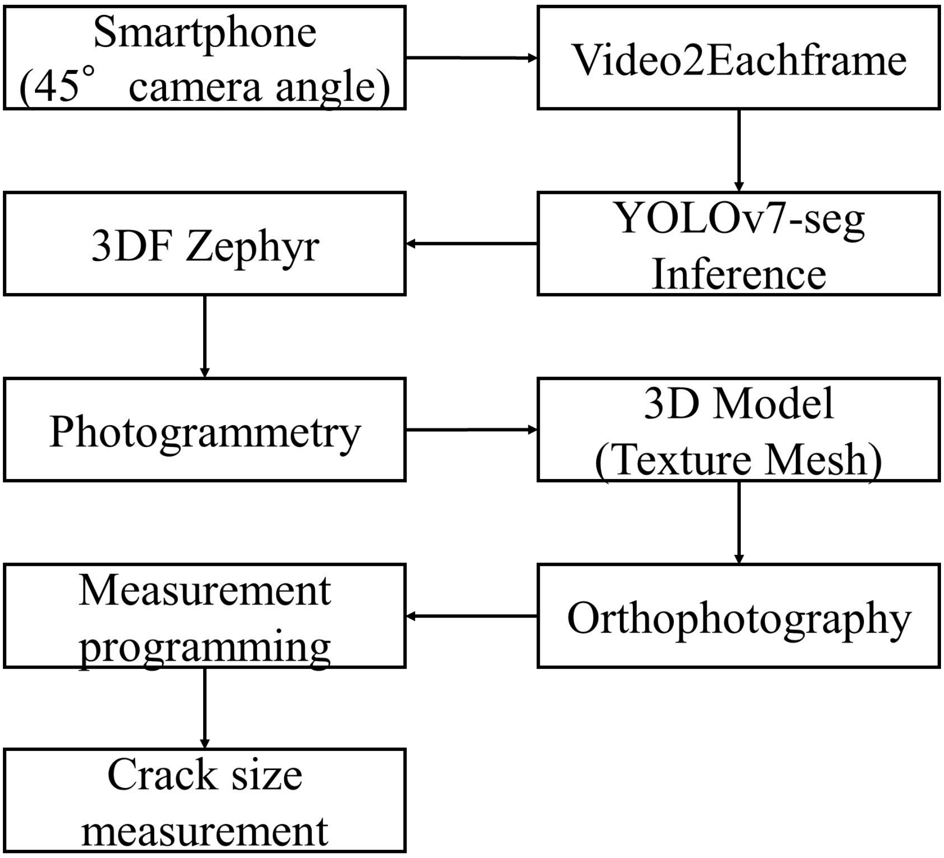

Figure Table2 Process of the data collection and the workflow.

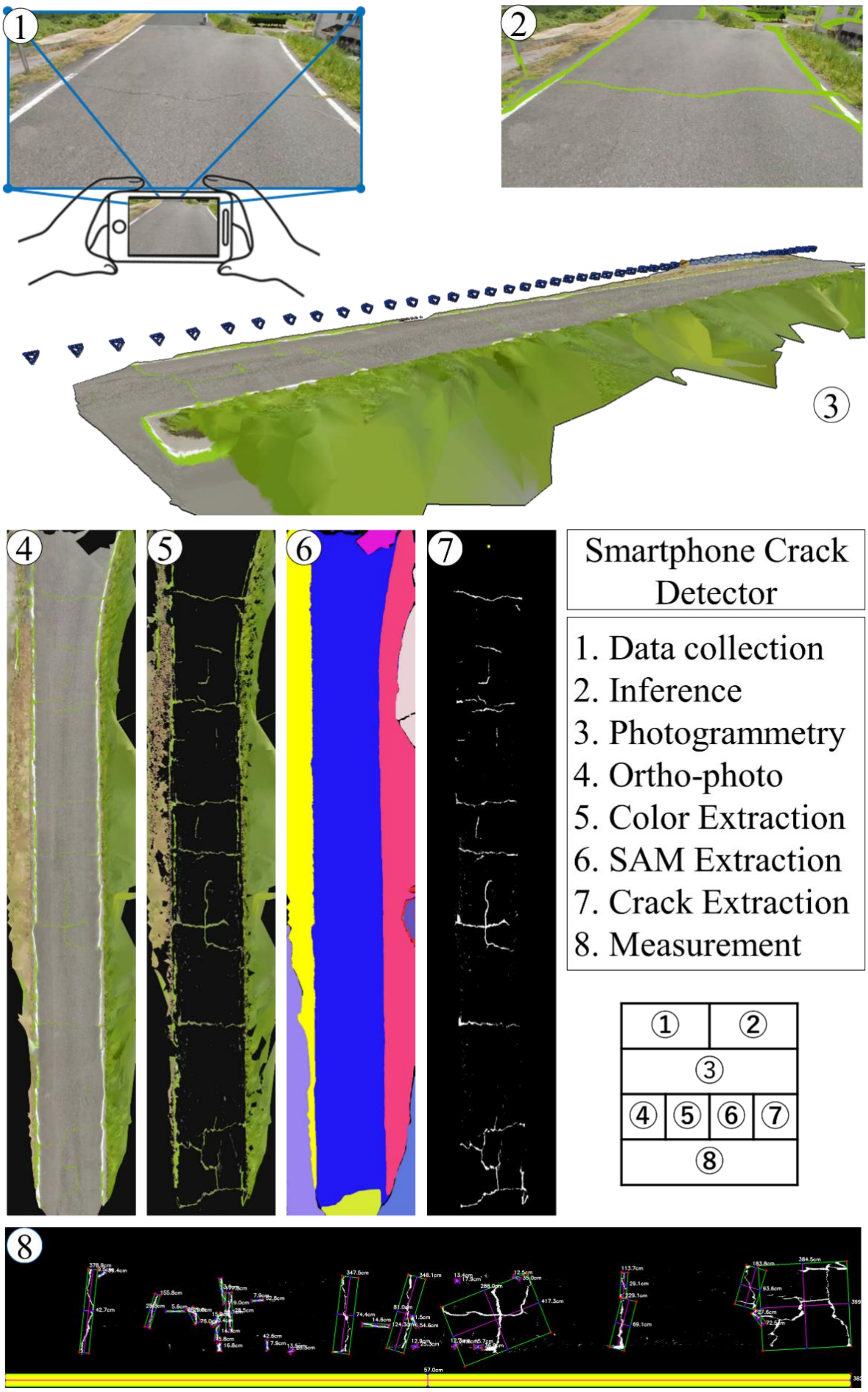

Figure 3 Process of the crack size measurement: Derived from the 3D Model Processing (i.e., Step-1 to Step-3), Ortho-photo in Step-4 was filtering by color extraction firstly (i.e., the YOLOv7-seg segmented mask color in Step-5), then extracting the asphalt-paved road from Step-5 using SAM (i.e., Segment Anything Model) in Step-6. Based on the mentioned steps, Step-7 extracted and recolor the cracks-/no-crack- parts. With the assistance of the reference object length using OpenCV programming in Step-8, the size of the cracks can be measured.

3 Work flow

A systematic process for measuring crack size using various technologies and methods is outlined in the workflow shown in the Figure 2 and 3. It starts with the acquisition of videos using a smartphone shown in the Figure 4 with a fixed camera angle of around 45°. These videos are then processed by Video2Eachframe.py (i.e. a python program to crop the video into multiple frames by setting) to be analyzed by YOLOv7-seg Inference using the parameters in Table 1. Simultaneously, the images are used to create a 3D texture mesh as shown in Figure 5, which is further refined by photogrammetry using 3DF Zephyr Free (i.e., a software that rebuild the 3D model from specified images shown in Figure 6, 50 images in this research). Orthophoto derived from the 3D Model is used along with measurement programming to facilitate crack size measurements. The final result in Figure 3, provides detailed cracked measurements. If necessary, as depicted in Figure 7, the mesh-based crack extraction is also possible.

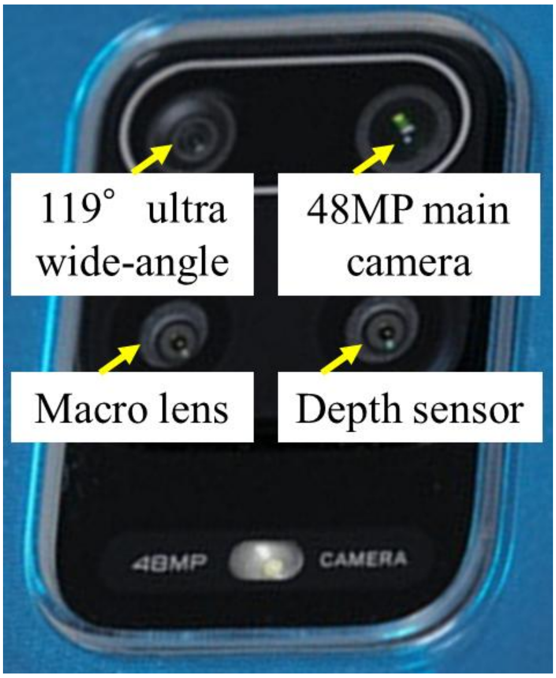

Figure 4 The camera sensors that were used for collecting the videos.

Table 1 Computer-related parameter setting.

	Train (Inference)
GPU memory (GB)	23 (2.8)
Batch size	12
Epochs	1000
Optimizer	SGD
Lr0	0.01
Image size (pixel)	1024 (1200)

Figure 5 Process of the generating 3D Model (Texture Mesh): Step-1, 50 selected video-derived frames in the project list; Step-2, extracting the sparse point cloud from the frames; Step-3, extracting the density point cloud from the frames; Step-4, constructing the mesh from the point cloud; Step-5, constructing the texture mesh and cropping the necessary parts.

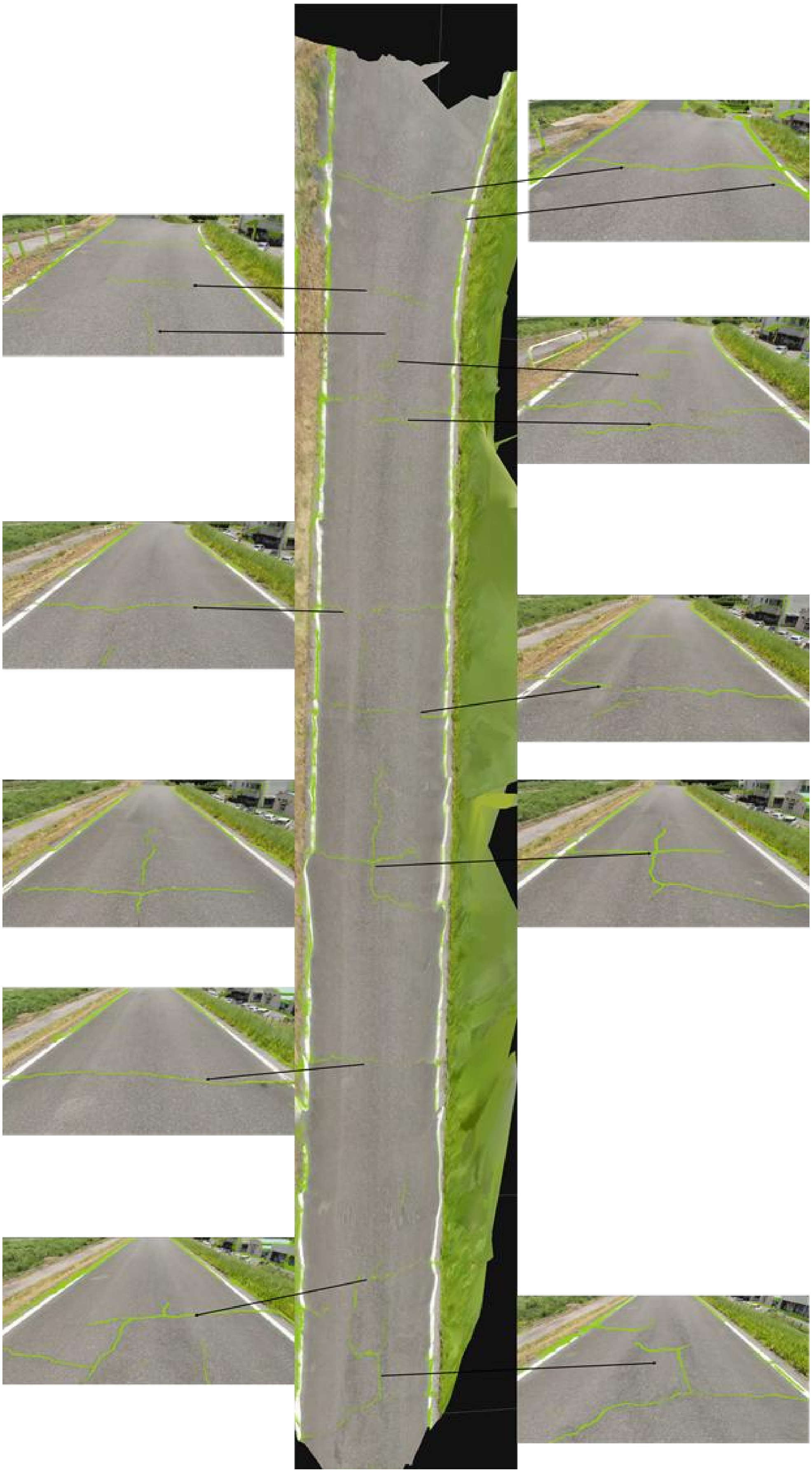

Figure 6 Samples of the specified images with the asphalt-paved crack.

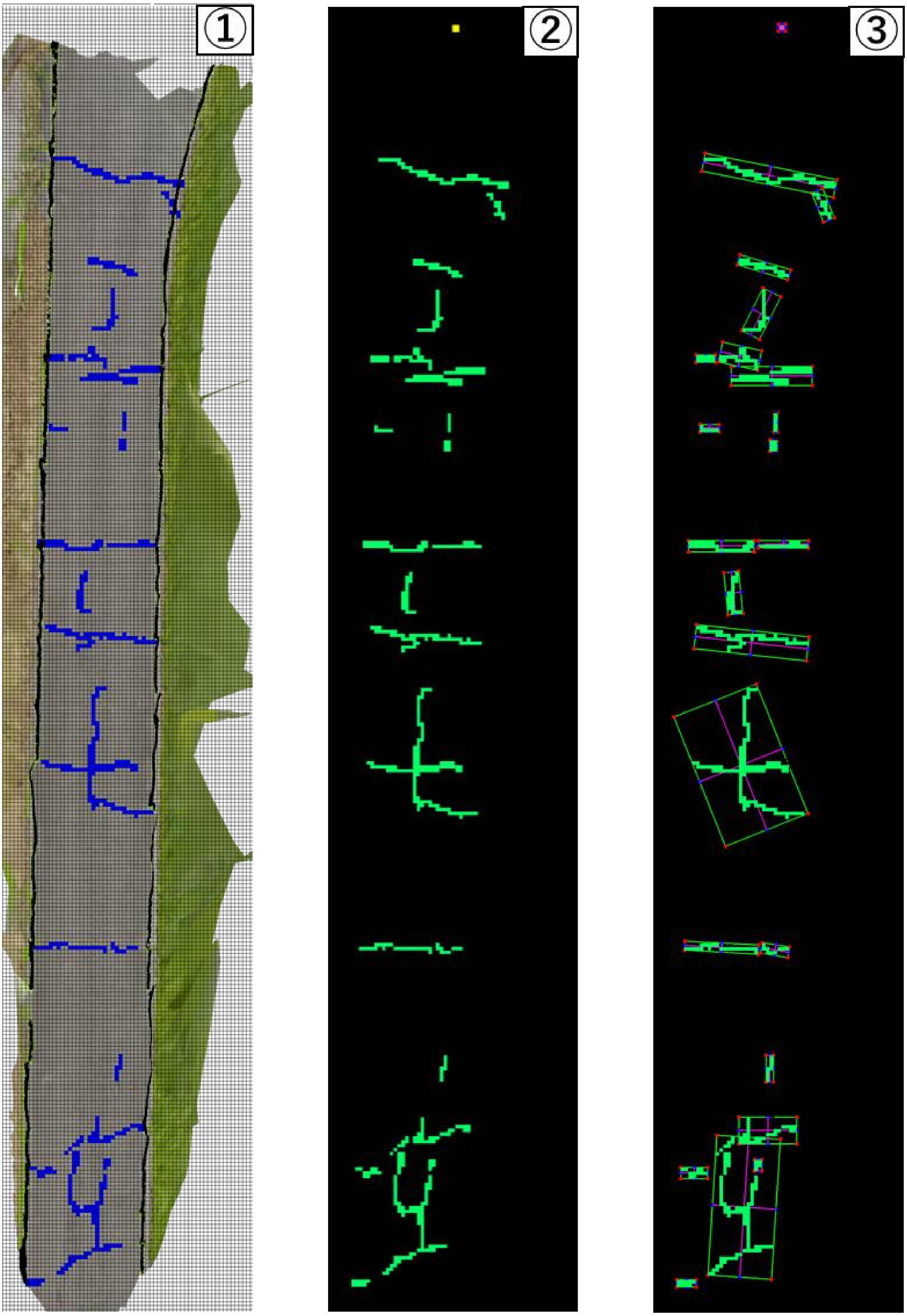

Figure 7 Mesh-based crack extraction.

4 Results

From the results shown in the Figure 3 and 7, to some degree, the angle-fixed video-only input can support the river area environmental management personnel to extract the riparian asphalt-paved road cracks under the situation of knowing the walking distance.

5 Conclusion

This study has extracted the cracks with the detailed information from the video after the mentioned process. In this process, orthophoto ensures accurate surface representation, while measurement programming facilitates the crack size determination. This comprehensive approach yields detailed measurements, enhancing our understanding of cracks and aiding in maintenance and safety assessments.

Reference

[1] Shijun, P. A. N., YOSHIDA, K., & NISHIYAMA, S. Detection and Segmentation of Riparian Asphalt Paved Cracks Using Drone and Computer Vision Algorithms. Intelligence, Informatics and Infrastructure, 4(2), 35-49. 2023